

SOLPS-ITER download and installation instructions

Welcome to SOLPS-ITER!

You can generally browse the ITER Git repositories here: <https://git.iter.org> and SOLPS-ITER specifically here: <https://git.iter.org/projects/BND/repos/solps-iter/browse>, but since these repositories are restricted you will first need to authenticate with your ITER username and password – possibly twice, once for the intranet, and once for the Git web front end in the top right-hand corner.

If you upload a copy of your public ssh key here:

<https://git.iter.org/plugins/servlet/ssh/account/keys> (by following links under Manage Account), then you should be able to simply clone a copy of SOLPS-ITER to your local machine by doing:

```
> git clone ssh://git@git.iter.org/bnd/solps-iter.git
> cd solps-iter
> git submodule init
> git submodule update
```

Since your userid is immutably encoded into any changes, it's always a good idea to properly configure your local installation of Git on all the systems you plan on making code revisions on with, e.g.:

```
> git config --global user.name "Your Name"
> git config --global user.email your.name@where.you.are
```

When updating the code, if you have a local branch where you have done some local work (or if your local work is not yet on a branch), make sure to commit or stash it first, and then you will be able to rebase it on top of the updated code as necessary. Please check the README file in your \$SOLPSTOP directory for instructions on how to safeguard your work and update your code version.

When you recompile the updated code, it is recommended that you to start with:

```
> gmake listobj
> gmake depend
> gmake {your favourite target[s] here}
```

as the location and list of dependencies for some files may have changed.

All users have write access to these repositories although not to the develop, master or svn/* branches. This basically means that, for your own development, if any, you should create \$USERNAME/{feature/[topic]|bugfix/[topic]|hotfix/[topic]} branches and raise a Pull Request (<https://git.iter.org/projects/BND/repos/solps-iter/pull-requests>) to have them merged into the develop branch after pushing them back to the ITER repository, where \$USERNAME is your ITER IDM account name. This naming convention is meant to allow the SOLPS-ITER ROs to more easily manage everybody's branches. Please conform to it.

Issues are handled using JIRA (<https://jira.iter.org/browse/IMAS>) and you should feel free to create new issues associated with SOLPS-ITER or any of its submodules (selected from the drop-down list of components) if you have any problems or suggestions. (You can ignore the options associated with resource estimates and sprints.)

For help with Git, we can recommend the documentation here (<http://git-scm.com/>) and the Pro Git book (linked from there) in particular.

The documentation specific to SOLPS-ITER can be found at:

<https://portal.iter.org/departments/POP/CM/IMAS/SOLPS-ITER>

and it is visible from here:

<https://imas.iter.org/> (then the “*Science Division*” tab on the left, and choose “*Confinement & Modelling*” from the drop-down list, then follow the link to the SOLPS-ITER Folder).

(where you will also find a manual on how to raise JIRA issues and pull requests and the like).

Once you have cloned the SOLPS-ITER Repository, you may need to modify the configuration files needed so the necessary environment variables are properly defined. Please check the README file and the SOLPS-ITER User Manual (you can use the documentation link above) for details. These include your domain hostname, defined with the “whereami” script, your preferred compiler, in the “default_compiler” script, the modules you wish to load and your preferred debugger name (given with the DBX=debugger_name command) in SETUP/setup.csh.\$HOST_NAME.\$COMPIILER, and the location and linking instructions for the various libraries in SETUP/config.\$HOST_NAME.\$COMPIILER. Feel free to look at the examples already present for inspiration. For hosts and compilers on which SOLPS-ITER has already been installed successfully and we know about, we have tried to include these files already in the distribution, but they may be mistaken or incomplete, so please check.

Once these variables are set, to start a SOLPS-ITER session (every time), from the SOLPS-ITER top directory, you do:

```
> tcsh
> source setup.csh
```

Then, if this is the first installation in your file space, please do:

```
> first_setup
```

This will, if they are not already present in the distribution, create the various config files for each of the SOLPS-ITER submodules. Feel free to modify them afterwards according to your local requirements. We strongly recommend that you do any modifications you may need in local copies of these files, to be named, respectively: SETUP/setup.csh.\$HOST_NAME.\$COMPIILER.local and SETUP/config.\$HOST_NAME.\$COMPIILER.local. Once you have made the proper modifications, if these are worth sharing with other users on similar platforms, please copy them over into the original files and commit them back to the repository using a pull request.

Then, if all was filled in properly, a simple “gmake” compilation command (and a LOT of patience) will get you a ready-to-run code. You should run the compilation from the SOLPS-ITER top directory, because this is where all the compiler flags are defined and passed down to each of the submodules. The top directory Makefile will also build the SOLPS-ITER Manual, which will be placed in SOLPS-ITER/doc/solp/solps.[pdf|dvi|ps].

For the libraries required at compilation, we recommend these be loaded as modules, which can then be shared across multiple users on a single machine, to make the SOLPS-ITER distribution

lighter. With the exception of the MSCL and libsonnet libraries, the libraries are not part of the SOLPS-ITER repository, as the IO does not have distribution rights for them. Libsonnet is built automatically from the SOLPS-ITER Makefile (although you may need to provide a machine architecture environment variable value). MSCL can be obtained from the ITER GIT server at:

<https://git.iter.org/projects/LIB/repos/mscl/browse>

When compiling the MSCL library, you will need to set two environment variables, in the following way (provided you have already loaded the SOLPS-ITER environment as above):

```
> setenv OBJECTCODE $COMPILER  
> setenv SOLPS_LIB $SOLPSLIB
```

For the others, most are freely available from the Internet, with the exception of the NAG library, for which a pay licence is required. However, the NAG library is not strictly necessary to build SOLPS-ITER. And, if you already have an older version of SOLPS (5.x or 4.x), as distributed by IPP-Garching, you may copy the libraries from it into the SOLPS-ITER/lib/\$HOST_NAME.\$COMPILER directory.

If you have any questions about these instructions or encounter difficulties, feel free to contact the SOLPS-ITER RO at:

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